

Linear first order diff eq: $p(x)y' + q(x)y = h(x)$
 $y' + a(x)y = f(x)$

$y_1 \rightarrow y' + a(x)y = 0$ (Homogeneous M)

$$u y_1 \quad u y_1' + u' y_1 + a(x) u y_1 = f(x)$$

$$u' y_1 = f(x)$$

Theorem (Existence and Uniqueness of solution: $y' = f(x, y)$)

Let $D = (a, b) \times (c, d)$ be an open rectangle containing the point (x_0, y_0) and consider the IVP

$$y' = f(x, y), \quad y(x_0) = y_0$$

(a) (Existence) Assume $f(x, y)$ is continuous on D . Then IVP has at least one solution on some interval $(a_1, b_1) \subset (a, b)$ containing x_0 .

(b) (Uniqueness) If both $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous on D , then IVP has a unique solution on some interval $(a', b') \subset (a, b)$ containing x_0 .

(c) (Interval of validity) Given a solution of the IVP, the largest interval containing x_0 on which the solution is defined is called the interval of validity of the solution.

Theorem

Consider ODE $M(x, y) + N(x, y) \frac{dy}{dx} = 0$.

Assume functions $M, N, \frac{dM}{dy}$ and $\frac{dN}{dx}$ be continuous in an open rectangle $R := \{a < x < b, c < y < d\}$.

Then the above ODE is exact if and only if M and N satisfies the condition $\frac{dM}{dy} = \frac{dN}{dx}$.

In other words, there exists a function $G: R \rightarrow \mathbb{R}$ such that $\frac{dG}{dx} = M$ and $\frac{dG}{dy} = N$ if and only if $\frac{dM}{dy} = \frac{dN}{dx}$ on R .

Bernoulli: $y' + p(x)y = f(x)y^r$
 (Homogeneous M)

Any non linear ODE $y' = q(y/x)$ can be converted to a separable equation by substituting $y = vx$.

Example

Solve IVP

$$y' = 2xy^2, \quad y(0) = y_0$$

and find the interval of validity.

A family of solutions is

$$y = \frac{-1}{x^2 + C}$$

- If $y_0 = 0$, the solution is $y \equiv 0$ and the interval of validity is $\mathbb{R}$.
- If $y_0 \neq 0$, then $C = -\frac{1}{y_0}$. Hence $y = \frac{-y_0}{y_0 x^2 - 1}$.
- If $y_0 < 0$, the solution is defined for all x . Hence the interval of validity is $\mathbb{R}$.
- If $y_0 > 0$, the solution is valid when $x \in \mathbb{R} - \{\pm 1/\sqrt{y_0}\}$.
Hence the interval of validity is $\left(\frac{-1}{\sqrt{y_0}}, \frac{1}{\sqrt{y_0}}\right)$.

Example (continued ...)

Thus the IVP $x^2 y' = y^2 + xy - x^2, \quad y(0) = 0$ has infinitely many solutions

$$y(x) = \begin{cases} x \frac{1 + C_1 x^2}{1 - C_1 x^2} & \text{if } x < 0 \\ x \frac{1 + C_2 x^2}{1 - C_2 x^2} & \text{if } x \geq 0 \end{cases}$$

one for each choice of C_1, C_2 .

The interval of validity I of $y(x)$ depends on C_1, C_2 .

- If $C_1 \leq 0$ and $C_2 \leq 0$, then $I = \mathbb{R}$.
- If $C_1 \leq 0$ and $C_2 > 0$, then $I = (-\infty, 1/\sqrt{C_2})$.
- If $C_1 > 0$ and $C_2 \leq 0$, then $I = (-1/\sqrt{C_1}, \infty)$.
- If $C_1 > 0$ and $C_2 > 0$, then $I = (-1/\sqrt{C_1}, 1/\sqrt{C_2})$.

Definition

We say $\mu(x, y)$ is an integrating factor of ODE

$$M(x, y) + N(x, y)y' = 0$$

if

$$\mu M + \mu N y' = 0$$

is exact, i.e.

$$\frac{d\mu}{dy} M + \mu \frac{dM}{dy} = \frac{d\mu}{dx} N + \mu \frac{dN}{dx}$$

that is,

$$\mu(M_y - N_x) = \frac{d\mu}{dx} N - \frac{d\mu}{dy} M$$

If we assume that $\mu(x, y) = P(x)Q(y)$, then

$$\mu(M_y - N_x) = \frac{d\mu}{dx} N - \frac{d\mu}{dy} M$$

$$\Rightarrow P(x)Q(y)(M_y - N_x) = P'(x)Q(y)N - P(x)Q'(y)M$$

$$\Rightarrow M_y - N_x = \frac{P'}{P}N - \frac{Q'}{Q}M \quad (\text{check for } p(x)=1 \text{ or } q(x)=1) \text{ (or)}$$

If

$$M_y - N_x = p(x)N - q(y)M \quad (\text{Try for } p \text{ or } q \text{ some } p \neq 2 \text{ or } p=0 \text{ or } q=0)$$

$$\frac{P'}{P} = p(x), \quad \frac{Q'}{Q} = q(y)$$

$$\Rightarrow P(x) = e^{\int p(x) dx}, \quad Q(y) = e^{\int q(y) dy}$$

Then

$$\mu(x, y) = e^{\int p(x) dx} e^{\int q(y) dy}$$

is an integrating factor.

Picard's Iteration Method:

$$y' = f(t, y) ; \quad y(a) = y_0$$

$$y_0(t) = y_0$$

$$y_1(t) = \int_a^t f(t, y_0(t)) dt + y_0$$

$$y_2(t) = \int_a^t f(t, y_1(t)) dt + y_0$$

$\vdots$

$$y_{n+1}(t) = \int_a^t f(t, y_n(t)) dt + y_0$$

$$y_{n+1}(t) = y_n(t) \quad , \text{ series converges}$$

$$y_n(t) = \int_a^t f(t, y_n(t)) dt + y_0$$

2nd Order:

Constant coeff equation : $ay'' + by' + cy = F(x)$

Theorem

Let $p(m) = am^2 + bm + c$ be the characteristic polynomial of

$ay'' + by' + cy = 0$, where $a, b, c \in \mathbb{R}$, $a \neq 0$. Then

- ① If $p(m) = 0$ has distinct real roots m_1, m_2 , then the general solution is $y(x) = c_1 e^{m_1 x} + c_2 e^{m_2 x}$
- ② If $p(m) = 0$ has repeated real roots m_1, m_1 , then the general solution is $y(x) = e^{m_1 x} (c_1 + c_2 x)$
- ③ If $p(m) = 0$ has complex conjugate roots $m_1 = \lambda + i\omega$ and $m_2 = \lambda - i\omega$, where $\omega > 0$, then the general solution is

$$y(x) = e^{\lambda x} [c_1 \cos(\omega x) + c_2 \sin(\omega x)]$$

Theorem

Consider the IVP

$$y'' + p(x)y' + q(x)y = 0, \quad y(x_0) = a, y'(x_0) = b,$$

where $p(x)$ and $q(x)$ are continuous on an interval I . Then there is a unique solution to the IVP on I .

Theorem (Dimension Theorem)

If $p(x), q(x)$ are continuous on an open interval I , then the set of solutions of the ODE

$$y'' + p(x)y' + q(x)y = 0 \quad (1)$$

on I is a vector space of dimension 2. Any basis $\{y_1, y_2\}$ of solutions of (1) is called a **fundamental set of solutions** of (1).

Theorem

Let p, q be continuous on some open interval I .

If $y_1 \neq 0$ is a solution of

$$y'' + p(x)y' + q(x)y = 0$$

on I , then another solution of ODE on I is given by

$$\begin{aligned} y_2(x) &= v y_1(x) \\ &= \left(\int \frac{e^{-\int p dx}}{y_1^2} dx \right) y_1(x) \end{aligned}$$

Further, y_1 and y_2 are linearly independent on I .

Theorem (Abel's Formula)

Assume $p(x)$ and $q(x)$ are continuous on $I = (a, b)$. Let $f(x)$ and $g(x)$ be solutions of $y'' + p(x)y' + q(x)y = 0$.

Then Wronskian of $f(x)$ and $g(x)$ is given by

$$W(f, g; x) = W(f, g; x_0) e^{-\int_{x_0}^x p(t) dt},$$

for any $x_0 \in I$.

Theorem

Consider

$$y'' + p(x)y' + q(x)y = 0,$$

where $p(x)$ and $q(x)$ are continuous on $I = (a, b)$. Suppose f and g are solutions on I . Then f and g are linearly independent on I if and only if $W(f, g; x)$ has no zeros in I .

Cauchy - Euler:

$$x^2 y'' + axy' + by = 0$$

$$x = e^t \quad \frac{dy}{dx} = \frac{dy}{dt} \frac{1}{e^t}$$

$$\frac{d^2 y}{dt^2} \frac{1}{e^t} - \frac{dy}{dt} e^{-2t} = \frac{d^2 y}{dx^2}$$

$$y'' - y' + ay' + by = 0$$

$$y'' + (a-1)y' + by = 0$$

Theorem

Consider the Cauchy-Euler Equation

$$x^2 y'' + axy' + by = 0, \quad x > 0 \quad (*)$$

Substituting $x = e^t$, ODE becomes

$$y''(t) + (a-1)y'(t) + by(t) = 0 \quad (**)$$

Let m_1 and m_2 be the roots of the char equation of (**)

$$p(m) = m^2 + (a-1)m + b = 0.$$

Then the general solution of (*) is given as follows.

- ① If $m_1 \neq m_2 \in \mathbb{R}$, then $y(x) = c_1 x^{m_1} + c_2 x^{m_2}$.
- ② If $m_1 = m_2 \in \mathbb{R}$, $y(x) = c_1 x^m + c_2 x^m \ln x$.
- ③ If $m_i = \lambda \pm i\omega$ with $\omega > 0$, then

$$y(x) = c_1 x^\lambda \cos(\omega \ln x) + c_2 x^\lambda \sin(\omega \ln x).$$

Non-homo 2nd order linear : Variation of parameters

$$y'' + p(x)y' + q(x)y = r(x)$$

y_1, y_2 independent solns of homo

$$\text{particular soln : } y = v_1 y_1 + v_2 y_2$$

$$\& v_1' y_1 + v_2' y_2 = 0$$

$$y' = v_1 y_1' + v_2 y_2'$$

$$y'' = v_1' y_1' + v_1 y_1'' + v_2' y_2' + v_2 y_2''$$

$$v_1' y_1' + v_2' y_2' = r(x)$$

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} v_1' \\ v_2' \end{bmatrix} = \begin{bmatrix} 0 \\ r(x) \end{bmatrix}$$

- (5) Let $Ly = (1-x)xy'' + 2xy' - 2y$. Let y_1, y_2 be linearly independent solutions of $Ly = 0$, $0 < x < 1$ with $y_1(x) = x, 0 < x < 1$; $y_2(\frac{1}{2}) = -\frac{3}{4} + \ln 2$, $y_2'(\frac{1}{2}) = 2 \ln 2 - 1$. If $W(x)$ denotes the wronskian of (y_1, y_2) in that order, and $y_p(x) = v_1(x)y_1(x) + v_2(x)y_2(x)$ is a particular solution of $Ly = x^2(1-x)^3$, then

$$y_2(x) =$$

$$W(0.5) =$$

$$v_1(x) =$$

$$v_2(x) =$$

$$\begin{aligned} xu \\ xu' + u \\ xu'' + 2u' \end{aligned}$$

$$(1-x)x^2u'' + 2(1-x)xu' + 2x^2u' - 2xu - 2xu = 0$$

$$(x-1)xu'' = 2u'$$

$$\frac{u''}{u'} = \frac{2}{(x-1)x}$$

$$u' = \frac{(x-1)^2}{x^2} + C$$

$$x^2 - 2 \ln x - 1$$

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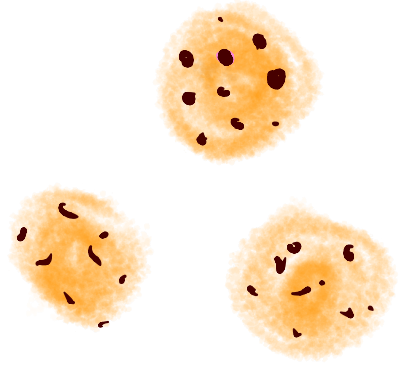
Sanaad Sabassum



ice cream



cinnamon roll



cookies



cup cake!



Aradhand's
bday cake



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n -order linear diff eqn $\leftarrow \begin{cases} y^{(n)} + p_1(x)y^{(n-1)} + p_2(x)y^{(n-2)} + \dots + p_n(x)y = r(x) \end{cases}$

$$\text{IVP} \begin{cases} y^{(n)} + p_1(x)y^{(n-1)} + p_2(x)y^{(n-2)} + \dots + p_n(x)y = 0 \\ y(x_0) = k_0 \\ y'(x_0) = k_1 \\ \vdots \\ y^{(n-1)}(x_0) = k_{n-1} \end{cases}$$

If p_i is cont on interval I where $x_0 \in I$,
IVP has unique soln on I

Wronskian:

$$C^n(I) = \{f: I \rightarrow \mathbb{R} \mid f, f', \dots, f^{(n)} \text{ are cont}\}$$

Let $f_1, f_2, \dots, f_n \in C^n(I)$, their wronskian = $W(f_1, f_2, \dots, f_n; x)$

$$= \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1^{(1)} & f_2^{(1)} & \dots & f_n^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$$

Abel's thm:

$$L = D^n + p_1(x)D^{n-1} + p_2(x)D^{n-2} + \dots + p_n(x)$$

$p_1, p_2, \dots, p_n$ are continuous in I

$y_1, y_2, \dots, y_n$ are the solutions of $Ly = 0$

$x_0 \in I$, Wronskian of $\{y_1, \dots, y_n\}$ is $W(x) = W(x_0) e^{-\int_{x_0}^x p_1(t) dt}$ where $x \in I$

So, W has either no zeros of I or $W \equiv 0$ on I

Equivalent statements:

- 1) $\{f_1, f_2, \dots, f_n\}$ fundamental set of solns of $Ly = 0$ on I
- 2) $f_1, f_2, \dots, f_n$ linearly independent on I
- 3) W is non zero at some point in I
- 4) W " " " " for whole interval I

Dimension Thm:

$$L = D^n + p_1(x) D^{n-1} + p_2(x) D^{n-2} + \dots + p_n(x) I$$

where p_i 's are cont

Dimension of vector space of sols to $Ly=0$ is n

Proof:

Let $(y_i(x_0), y_i^{(1)}(x_0), \dots, y_i^{(n-1)}(x_0)) = e_i$ be the standard basis
 $x_0 \in I$

$$Ly=0, e_i \longrightarrow \text{IVP}$$

Let $y_1, y_2, \dots, y_n$ be sols to the IVP

$$W(y_1, y_2, \dots, y_n; x) = 1 \Rightarrow y_1, y_2, \dots, y_n \text{ are linearly independent}$$

Let y is any soln to $Ly=0$

$$\text{Let } z(x) = y(x_0) y_1(x) + y^{(1)}(x_0) y_2(x) + \dots + y^{(n-1)}(x_0) y_n(x)$$

$$z(x_0) = y(x_0)$$

$$z'(x_0) = y'(x_0)$$

$\vdots$

$$\text{so, } y(x) = z(x) \text{ on } x \in I$$

Constant coeff diff operator:

$$L = a_0 D^{(n)} + a_1 D^{(n-1)} + \dots + a_n I$$

$$\text{If } L = \sum_{i=0}^n a_i D^{n-i} \text{ and } M = \sum_{j=0}^m b_j D^{m-j}, L(M(f)) = M(L(f)) \quad \forall f \in C^{m+n}$$

$L, M \rightarrow$ const coeff diff operators

$$\text{Eg: } y''' - 7y' - 6y = 0$$

$$D^3 - 7D - 6 = 0$$

$$(D+1)(D+2)(D-3) = 0$$

$$e^{-x} \quad e^{-2x} \quad e^{3x}$$

$$y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{3x} + c_f$$

To check if linearly independent, $W(e^{-x}, e^{-2x}, e^{3x}; 0) \neq 0$
 in any interval containing $x=0$

$\text{Ker}(L) \rightarrow$ space of funcs y st $Ly=0$ where L is const coeff op.

$$e^{rx}, x e^{rx}, \dots, x^{m-1} e^{rx} \in \text{Ker}((D-r)^m);$$

$$e^{ax} \sin bx, e^{ax} \cos bx, \in \text{Ker}((D-a)^2 + b^2)^m$$

$$x e^{ax} \sin bx, x e^{ax} \cos bx,$$

$\vdots$

$$x^{m-1} e^{ax} \sin bx, x^{m-1} e^{ax} \cos bx$$

Annihilator or Undetermined Coeff Method:

- To find particular soln of const coeff non-homo diff eqn.
 $Ly = r(x)$
- Find diff op M s.t $M(r(x)) = 0$
- Find basis of $MLy = 0$, exclude the basis set of $Ly = 0$ from the $MLy = 0$ basis set, let the linear combination of the remaining basis set elements be y_p
- Solve for the coeff of basis set ele of y_p thru $Ly_p = r(x)$ and substitute in y_p
- Gen soln for $Ly = r(x)$ is $y_p + z$, where z is the general soln of $Ly = 0$

Variation of Parameters:

$$y^{(n)} + p_1(x)y^{(n-1)} + p_2(x)y^{(n-2)} + \dots + p_n(x)y = r(x)$$

$y_1, y_2, \dots, y_n \rightarrow$ basis of solns of homogeneous part

$$y_p = v_1(x)y_1 + v_2(x)y_2 + \dots + v_n(x)y_n$$

$$\text{Let } v_1' y_1 + v_2' y_2 + \dots + v_n' y_n = 0,$$

$$v_1' y_1' + v_2' y_2' + \dots + v_n' y_n' = 0,$$

$$\vdots$$

$$v_1' y_1^{(n-2)} + v_2' y_2^{(n-2)} + \dots + v_n' y_n^{(n-2)} = 0$$

$$\text{we get, } v_1' y_1^{(n-1)} + v_2' y_2^{(n-1)} + \dots + v_n' y_n^{(n-1)} = r(x) \quad (\text{after putting } y_p \text{ in eqn})$$

$$\begin{bmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{bmatrix} \begin{bmatrix} v_1' \\ v_2' \\ \vdots \\ v_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ r(x) \end{bmatrix}$$

$$v_i' = \frac{\begin{vmatrix} y_1 & \dots & y_{i-1} & 0 & y_{i+1} & \dots & y_n \\ y_1' & \dots & y_{i-1}' & 0 & y_{i+1}' & \dots & y_n' \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & \dots & y_{i-1}^{(n-1)} & r(x) & y_{i+1}^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix}}{W(y_1, y_2, \dots, y_n; x)}$$

$$y_p = v_1 y_1 + v_2 y_2 + \dots + v_n y_n$$

Improper Integral:

If g integrable in the interval $[a, T)$ for every $a < T$, then its improper integral of g over $[a, \infty)$

$$\int_a^{\infty} g(t) dt = \lim_{T \rightarrow \infty} \int_a^T g(t) dt$$

Laplace Transform:

Let f be defined for $t \geq 0$ & s be a real no.

Laplace Transform $F(s) = \int_0^{\infty} e^{-st} f(t) dt$ for values of s where it converges.

$$F = L(f) \quad f(t) \leftrightarrow F(s) \quad L(f(t)) = F(s)$$

Piece-wise Continuous func:

$$f: [0, T] \rightarrow \mathbb{R}$$

- $f(0^+)$, $f(T^-)$ exist & finite
- $0 \leq t_0 < t_1 < t_2 \dots < t_{n+1} = T$
 f on (t_{i-1}, t_i) $i \in \{1, \dots, n+1\}$ are cont
- $f(t_i^+)$, $f(t_i^-)$ exist & finite
 $i \in \{1, \dots, n\}$

$$\left(\frac{-1}{2}\right)! = \sqrt{\pi}$$

Exponential order:

f is piecewise cont func, exp order s_0

there exists $|f(t)| \leq M e^{s_0 t}$, $t \geq t_0$

If f is piecewise cont func in $[0, \infty)$, Laplace is defined for $s > s_0$.

Eg: $\sin \omega t$, $\cos \omega t \rightarrow s_0 = 0$

$L(\sin \omega t)$, $L(\cos \omega t)$ exists for $s > 0$

$$L(e^{at}) = \frac{1}{s-a} \quad s > a, a \in \mathbb{R}$$

$$L(te^{at}) = \frac{1}{(s-a)^2} \quad s > a$$

$$L(\sin wt) = \frac{w}{s^2 + w^2} \quad s > 0, w \in \mathbb{R}$$

$$L(\cos wt) = \frac{s}{s^2 + w^2} \quad s > 0$$

$$L(t^n) = \frac{n!}{s^{n+1}} \quad s > 0, n \geq 1$$

$$\cosh bt = \frac{s}{s^2 - b^2} \quad s > |b|$$

$$\sinh bt = \frac{b}{s^2 - b^2} \quad s > |b|$$

$$L(1) = \frac{1}{s} \quad s > 0$$

$$L(c_1 f_1 + \dots + c_n f_n) = c_1 L(f_1) + \dots + c_n L(f_n) \quad s > s_0 \downarrow \max(s_1, \dots, s_n)$$

$$L(f(t)) = F(s) \quad s > s_0$$

$$L(e^{at} f(t)) = F(s-a) \quad s > s_0 + a \quad (\text{First shifting thm})$$

$$L^{-1}(c_1 f_1 + \dots + c_n f_n) = c_1 L^{-1}(f_1) + \dots + c_n L^{-1}(f_n)$$

$$\frac{8+3s}{(s^2+1)(s^2+4)} = \frac{A+Bs}{s^2+1} + \frac{Cs+D}{s^2+4}$$

$$\begin{aligned} L(u(t-a)g(t)) &= e^{-as} L(g(t+a)) \quad (\text{Second shifting thm}) \\ \int_0^\infty u(t-a)g(t)dt &= \int_0^\infty g(t+a)e^{-st}e^{-sa}dt \\ &= e^{-sa} L(g(t+a)) \\ &\searrow \\ L(u(t-a)f(t-a)) &= e^{-as} L(f(t)) = e^{-as} F(s) \end{aligned}$$

Let f be cont on $[0, \infty)$ & of exp order s_0 , f' be piecewise cont on $[0, \infty)$

$$L(f') = sL(f) - f(0) \quad \text{for } s > s_0$$

We need initial conditions at $t=0$ to solve IVP.

$f, f', \dots, f^{(n-1)}$ cont on $[0, \infty)$ & exp order s_0

$f^{(n)}$ piecewise cont on $[0, \infty)$

$$L(f^{(n)}) = s^n L(f) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

Altho soln exists on $\mathbb{R}$, Laplace only gives soln from $[0, \infty)$

Unit or Heaviside Step func:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$$u(t-a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

$$f(t) = \begin{cases} f_1 & 0 \leq t < t_1 \\ f_2 & t_1 \leq t < t_2 \\ \vdots & \vdots \\ f_n & t_{n-1} \leq t \end{cases}$$

$$\begin{aligned} &= f_1 + (f_2 - f_1)u(t - t_1) + (f_3 - f_2)u(t - t_2) \\ &\quad + \dots + (f_n - f_{n-1})u(t - t_{n-1}) \end{aligned}$$

Find $L(f(t))$, where $f(t) = \begin{cases} 1, & 0 \leq t < 2 \\ -2t + 1, & 2 \leq t < 3 \\ 3t, & 3 \leq t < 5 \\ t - 1, & t \geq 5 \end{cases}$

$$f(t) = 1 - 2t(u(t-2)) + (5t-1)u(t-3) - (1+2t)u(t-5)$$

$$L(f(t)) = \frac{1}{s} - 2e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right) + e^{-3s}\left(\frac{5}{s^2} + \frac{14}{s}\right) - e^{-5s}\left(\frac{2}{s^2} + \frac{11}{s}\right)$$

$$F(s) = e^{-s} \frac{1}{2s} - e^{-2s} \frac{s+1}{(s+1)^2 + 1} \quad L^{-1}(F) = ?$$

$$\frac{1}{2} L^{-1}\left(\frac{e^{-s}}{s}\right) - L^{-1}\left(e^{-2s} \frac{s+1}{(s+1)^2 + 1}\right) = \frac{1}{2} u(t-1) - u(t-2) e^{-t+2} \sin(t-2)$$

$$L(f(t-a)u(t-a)) = e^{-as} L(f(t))$$

$$e^{-t}(\cos t)$$

$$L^{-1}(e^{-bs} F(s+a)) = u(t-b) e^{-a(t-b)} f(t-b)$$

$$L^{-1}(F(s+a)) = e^{-at} f(t)$$

$$y'' + y = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

$$f(t) = \begin{cases} 0, & 0 \leq t < \frac{\pi}{4} \\ \cos 2t, & \frac{\pi}{4} \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

$$f(t) = u\left(t - \frac{\pi}{4}\right) \cos 2t - \cos 2t u\left(t - \pi\right)$$

$$\cos\left(t - \frac{\pi}{4} + \frac{\pi}{4}\right) = \cos\left(2t - \pi\right) + 2\pi$$

$$L(f(t)) = -e^{-\frac{\pi}{4}s} \frac{2}{s^2 + 4} - e^{-\pi s} \frac{s}{s^2 + 4}$$

$$L(y) = \frac{-2e^{-\frac{\pi}{4}s}}{s} \left(\frac{1}{(s^2 + 4)} - \frac{2}{(s^2 + 4)} \right) - \frac{e^{-\pi s}}{s} \left(\frac{s}{(s^2 + 4)} - \frac{s}{(s^2 + 4)} \right)$$

$$y = -\frac{2}{s} u\left(t - \frac{\pi}{4}\right) \left(\sin\left(t - \frac{\pi}{4}\right) + \frac{1}{2} \sin\left(2t - \frac{\pi}{2}\right) \right)$$

$$- \frac{1}{s} u(t - \pi) \left(\cos\left(t - \pi\right) - \cos\left(2t - 2\pi\right) \right)$$

$$+ \frac{\cos t}{s} + \frac{\cos 2t}{s}$$

$$f(t) = \begin{cases} 0 & 0 \leq t < \frac{\pi}{4} \\ -\frac{1}{s} \cos 2t + \frac{\sqrt{2}}{s} (\cos t - \sin t) & \frac{\pi}{4} \leq t < \pi \\ \frac{1 + \sqrt{2}}{s} \cos t - \frac{\sqrt{2}}{s} \sin t & \pi \leq t \end{cases}$$

$$\int \cos at e^{bt} dt = \frac{e^{bt}}{a^2 + b^2} (a \sin at + b \cos at)$$

$$\int \sin at e^{bt} dt = \frac{e^{bt}}{a^2 + b^2} (b \sin at - a \cos at)$$

Convolution :

$$(f * g)(t) = \int_0^t f(z) g(t-z) dz$$

$$f * 1 \neq f$$

$$(f * g) * h = f * (g * h)$$

Convolution Thm :

If $L(f) = F(s)$ and $L(g) = G(s)$, the $L(f * g)$ exists, and

$$L(f * g) = L\left(\int_0^t f(z) g(t-z) dz\right) = F(s) G(s)$$

Proof :

$$\begin{aligned} & \int_0^{\infty} e^{-st} \int_0^t f(z) g(t-z) dz dt \\ &= \int_0^{\infty} f(z) \int_z^{\infty} e^{-st} g(t-z) dt dz \\ &= \int_0^{\infty} f(z) e^{-sz} \left(\int_0^{\infty} e^{-sx} g(x) dx \right) dz \\ &= F(s) G(s) \end{aligned} \quad \begin{aligned} \int_0^{\infty} e^{-st} f(t) dt &= F(s) \\ \int_0^{\infty} e^{-st} g(t) dt &= G(s) \end{aligned}$$

Give a formula for the solution of the IVP.

$$y'' + 2y' + 2y = f(t), \quad y(0) = a, \quad y'(0) = b$$

Taking Laplace transform gives,

$$(s^2 + 2s + 2)Y(s) = F(s) + b + as + 2a. \text{ Therefore,}$$

$$Y(s) = \frac{1}{s^2 + 2s + 2} F(s) + \frac{b + a + a(s + 1)}{s^2 + 2s + 2}$$

$$L^{-1}\left(\frac{1}{s^2 + 2s + 2}\right) = e^{-t} \sin t,$$

Hence

$$y(t) = \int_0^t f(t-\tau) e^{-\tau} \sin \tau d\tau + e^{-t} [(b+a) \sin t + a \cos t]$$

$$h(t) = \int_0^t x^m (t-x)^n dx$$

$$L(h(t)) = \frac{m!}{s^{m+1}} \frac{n!}{s^{n+1}}$$

$$h(t) = \frac{m! n!}{(m+n+1)!} t^{m+n+1}$$

$$h(t) = \int_0^t \sin a(t-z) \cos bz dz$$

$$L(h(t)) = \frac{a}{s^2 + a^2} \frac{bs}{s^2 + b^2}$$

$$h(t) = \frac{ab}{a^2 - b^2} (\cos bt - \cos at)$$

Volterra Integral Equation:

$$y(t) = f(t) + \int_0^t k(t-z)y(z) dz$$

Solve for y using convolution thm.

$$Y(s) = F(s) + K(s)Y(s)$$

$$Y(s) = \frac{F(s)}{1 - K(s)}$$

$$\rightarrow L(f * 1) = \frac{F(s)}{s}$$

Assume $L(f(t))$ is defined for $s > s_0$, then

- $L(e^{-at}f(t)) = F(s+a), \quad s > s_0 + a.$
- $L(u(t-a)f(t-a)) = e^{-as}F(s), \quad s > s_0, a > 0.$
- $L\left(\int_0^t f(\tau) d\tau\right) = \frac{F(s)}{s}, \quad s > \max\{0, s_0\}.$
- $L(tf(t)) = -F^{(1)}(s), \quad s > s_0.$
- $L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(s') ds', \quad s > s_0.$
- f : piecewise continuous and of exponential order. Then
 - (i) $\lim_{s \rightarrow \infty} F(s) = 0,$
 - (ii) $\lim_{s \rightarrow \infty} sF(s)$ is bounded.
- f, f' : piecewise continuous and of exponential order. Then $\lim_{s \rightarrow \infty} sF(s) = f(0).$
- If f is piecewise continuous and periodic of period T , then
$$L(f(t)) = \frac{1}{1 - e^{-sT}} \int_0^T f(t)e^{-st} dt, \quad s > 0$$

Properties:

$L(f(t))$ is defined for $s > s_0$

$$1) L(e^{at} f(t)) = F(s-a) \quad s > s_0 + a$$

$$2) L(u(t-a) f(t-a)) = e^{-as} F(s) \quad s > s_0, a > 0$$

$$3) L\left(\int_0^t f(z) dz\right) = \frac{F(s)}{s} \quad s > \max(0, s_0)$$

$$4) L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(r) dr \quad s > s_0$$

$$5) L\left(t^k f(t)\right) = (-1)^k F^{(k)}(s) \quad s > s_0$$

6) f be piecewise cont and of exp order.

$$\circ \lim_{s \rightarrow \infty} F(s) = 0$$

$$\circ \lim_{s \rightarrow \infty} s F(s) \text{ is bounded}$$

7) f, f' be piecewise cont and of exp order.

$$\lim_{s \rightarrow \infty} s F(s) = f(0)$$

8) f is piecewise cont & period T

$$L(f(t)) = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt \quad s > 0$$

Leibniz - Integral Rule:

$$F(x) = \int_{a(x)}^{b(x)} f(x, t) dt$$

$$\frac{d}{dx} (F(x)) = f(x, b(x)) b'(x) - f(x, a(x)) a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} (f(x, t)) dt$$

$$3) \quad L\left(\int_0^t f(z) dz\right) = \frac{F(s)}{s}$$

$$L(f * 1) = F(s) \left(\frac{1}{s}\right)$$

$$\begin{aligned} \text{Eg: } L^{-1}\left(\frac{1}{s^2(s^2+1)}\right) &= \int_0^t L^{-1}\left(\frac{1}{s(s^2+1)}\right) dz \\ &= \int_0^t \left(\int_0^z L^{-1}\left(\frac{1}{s^2+1}\right) dr\right) dz \\ &= \int_0^t \left(\int_0^z \sin r dr\right) dz \\ &= \int_0^t (1 - \cos z) dz \\ &= t - \sin t \end{aligned}$$

$$4) \quad L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(r) dr$$

$$\begin{aligned} \int_s^\infty \left(\int_0^\infty f(t) e^{-rt} dt\right) dr &= \int_0^\infty \left(\int_s^\infty f(t) e^{-rt} dr\right) dt \\ &= \int_0^\infty f(t) \int_s^\infty e^{-rt} dr dt \\ &= \int_0^\infty f(t) \frac{e^{-st}}{t} dt \\ &= L\left(\frac{f(t)}{t}\right) \end{aligned}$$